

Module 2: Data Link Layer & ARQ

Framing (Bit Stuffing), CRC-32 Division, Hamming Code, Stop-and-Wait, Go-Back-N & Selective Repeat

Module II: Transmission Media & Signal Encoding — Complete 6-Topic Tracker

- Topic 14:** Guided Media (Twisted Pair UTP/STP, Coaxial Cable, Optical Fiber)
- Topic 15:** Wireless Media & Propagation (Ground, Sky, LOS, Satellite Microwave)
- Topic 16:** Digital Signaling (Line Codes: NRZ-L, NRZ-I, Manchester, AML)
- Topic 17:** Analog Signaling (Carrier Parameter Representations)
- Topic 18:** Encoding Techniques (Line Coding Comparison & Tradeoffs)
- Topic 19:** Modulation Techniques (ASK, FSK, PSK & Constellation QAM)

Topic 14: Guided Transmission Media

Guided Medium	Physical Architecture & Transmission Physics	Key Advantages & Tradeoffs
1. Twisted Pair Cable	Two insulated copper wires twisted in a helical spiral. Twisting ensures both wires receive equal electromagnetic interference (EMI), which cancels out at the differential receiver. Available as UTP (Unshielded) and STP (Shielded with metal foil).	Inexpensive, lightweight, easy to terminate with RJ-45 connectors. Limited bandwidth (≤ 10 Gbps over short distances), prone to crosstalk.
2. Coaxial Cable	Solid central copper conductor surrounded by dielectric insulating layer, enclosed in a braided cylindrical wire mesh shield and outer protective plastic jacket.	Better noise immunity and higher bandwidth than twisted pair. Bulkier, stiffer, harder to install. Used in Cable TV broadband (DOCSIS).
3. Optical Fiber	Ultra-pure cylindrical glass core (n_1) surrounded by concentric glass cladding (n_2) where refractive index $n_1 > n_2$. Transmits light pulses via Total Internal Reflection whenever incident angle $\theta > \theta_{\text{critical}}$.	Enormous bandwidth (terabits/sec), ultra-low attenuation (≈ 0.2 dB/km), zero EMI susceptibility, lightweight, highly secure against wiretapping. Expensive splicing and optical transceivers.

Topic 15: Wireless Transmission & Propagation [UPLOADED PYQ]

Propagation Mode	Atmospheric Interaction & Frequency Band	Representative Applications
1. Ground-Wave Propagation	Signals follow the curvature of the earth via diffraction over the conductive ground surface (< 2 MHz).	AM radio broadcasting, maritime navigation.
2. Sky-Wave Propagation	Signals bounce back and forth between earth's surface and the charged ionosphere (2–30 MHz).	Shortwave (SW) international radio broadcasting, amateur radio.
3. Line-of-Sight (LOS)	Straight-line path between transmitting and receiving directional horn antennas (> 30 MHz). Antennas must have clear Fresnel zone.	Terrestrial microwave links, cellular base stations, Wi-Fi.

Satellite Microwave vs. Terrestrial Microwave [UPLOADED PYQ]:

Terrestrial Microwave: Uses parabolic dish antennas mounted on high towers spaced 30–50 km apart due to Earth's horizon curvature. Low propagation delay ($< 1\text{ ms}$).

Satellite Microwave: Earth station beams signal to a geostationary satellite at 35,786 km altitude (Uplink), which amplifies and retransmits it on a different downlink frequency. High round-trip latency ($\approx 270\text{--}540\text{ ms}$), but provides massive continental coverage.

Topic 16 & 18: Digital Line Coding Schemes [UPLOADED PYQ]

[UPLOADED PYQ] Step-by-Step Waveform Rules for Bit Stream '001110011'

Encoding Scheme	Bit Transition Invariant Rule	Waveform Encoding for '001110011'
NRZ-L (Level)	Bit '0' = Positive voltage ($+V$), Bit '1' = Negative voltage ($-V$).	' $+V, +V, -V, -V, -V, +V, +V, -V, -V$ '
NRZ-I (Invert)	Bit '1' = Transition at start of bit interval; Bit '0' = No transition.	Maintains previous level on '0', flips on '1'.
Bipolar AMI	Bit '0' = Zero voltage ($0V$); Bit '1' = Alternating $+V$ and $-V$.	' $0V, 0V, +V, -V, +V, 0V, 0V, -V, +V$ '
Manchester (802.3)	Bit '0' = High-to-Low transition at bit center; Bit '1' = Low-to-High transition at center.	Guarantees a mid-bit clock transition in every single bit!
Differential Manchester	Always transitions at center. Bit '0' = Transition at start; Bit '1' = No transition at start.	Token Ring standard; immune to wire polarity inversion.

Topic 17 & 19: Carrier Modulation Techniques (ASK, FSK, PSK, QAM)

Digital modulation alters one or more parameters of a high-frequency analog sinusoidal carrier $c(t) = A \cos(2\pi f_c t + \phi)$:

Modulation	Modulated Parameter	Mathematical Signal Equation	Key Advantages & Tradeoffs
ASK	Amplitude	$s(t) = A_1 \cos(2\pi f_c t)$ for '1', $A_0 \cos(2\pi f_c t)$ for '0'	Simple; highly susceptible to noise and fading.
FSK	Frequency	$s(t) = A \cos(2\pi f_1 t)$ for '1', $A \cos(2\pi f_2 t)$ for '0'	High noise immunity; requires more channel bandwidth.
BPSK / QPSK	Phase	$s(t) = A \cos(2\pi f_c t + \phi_i)$ where $\phi_i \in \{0, \pi\}$	Excellent bit error rate (BER); standard in modern wireless.
QAM (16/64/256)	Amplitude + Phase	$s(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$	Enormous spectral efficiency ($\log_2 M$ bits/ baud); used in Wi-Fi & 5G.



M2 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ] Draw the line coding waveforms for bit stream '001110011' using NRZ-L, NRZ-I, Bipolar AMI, Manchester, and Differential Manchester. (10 Marks)

1. **NRZ-L:** High for '0', Low for '1' $\rightarrow$ 'High, High, Low, Low, Low, High, High, Low, Low'.
2. **NRZ-I:** Invert signal level on every '1'; no change on '0'.
3. **Bipolar AMI:** Zero volts for all '0's; alternating $+V$ and $-V$ for each successive '1' $\rightarrow$ '0, 0, $+V$, $-V$, $+V$, 0, 0, $-V$, $+V$ '.
4. **Manchester:** Every bit has mid-interval transition. '0' = High-to-Low; '1' = Low-to-High.
5. **Differential Manchester:** Mandatory mid-bit transition. Start-of-bit transition for '0'; no start transition for '1'.